

ENCM 509 - Fundamentals of Biometric Systems Design
Laboratory Project #7: Face Recognition
Subproject 1 - LBP, SVM, and MLP Comparative Analysis
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1. Introduction

Face recognition is a biometric technique used to identify individuals based on facial features extracted from images. It is widely applied in security systems, access control, and identity verification. This project investigates face recognition using the AT&T (ORL) face database, comparing two classifiers, Support Vector Machine (SVM) and Multi-Layer Perceptron (MLP) with and without Local Binary Pattern (LBP) feature extraction.

The AT&T database consists of 400 grayscale images from 40 subjects, with 10 images per subject captured under varying lighting, expression, and pose. The dataset was split 50/50 into training and test sets, resulting in 200 training images and 200 test images with equal subject representation across both sets.

Local Binary Pattern (LBP) is a texture descriptor that encodes the local structure of an image by comparing each pixel to its neighbors and producing a binary code. These codes are summarized into a histogram that represents the texture of the face. LBP is computationally efficient and robust to illumination changes, making it well-suited for face recognition tasks.

Support Vector Machine (SVM) is a classifier that finds the optimal decision boundary separating classes in feature space. It is governed by the regularization parameter C and the kernel coefficient γ . Multi-Layer Perceptron (MLP) is a fully connected neural network that learns non-linear decision boundaries through backpropagation. Its main hyperparameters are the hidden layer architecture and the learning rate.

2. Procedure

2.1 Preprocessing

All images were loaded in grayscale and resized to 64x64 pixels. Pixel values were normalized to the range $[0, 1]$ by dividing by 255. This standardization ensures consistent input scale across all classifiers.

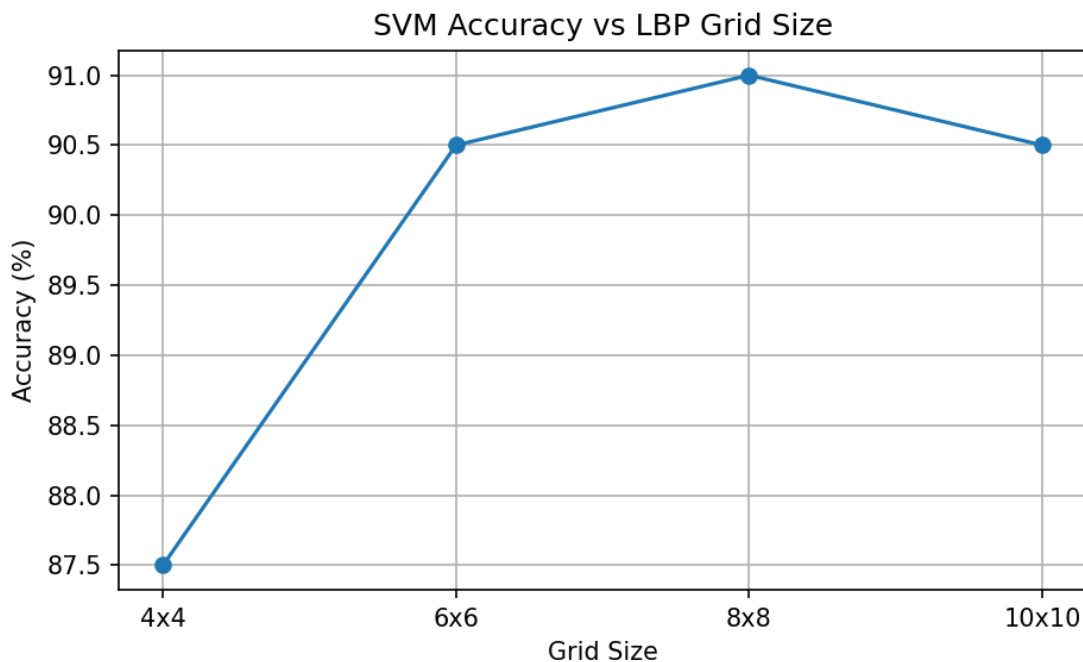
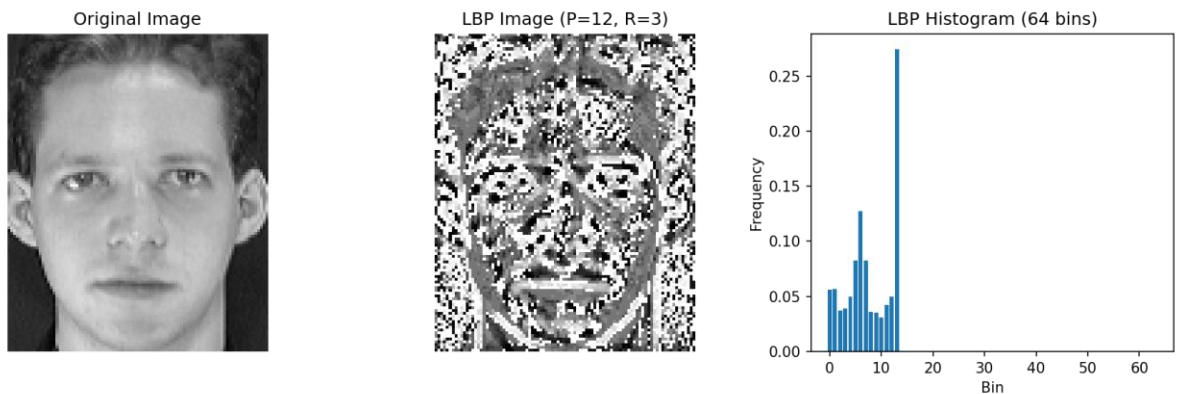
2.2 Feature Extraction

Two feature representations were used. The first is raw pixel features, where each 50x50 image is flattened into a 2500-dimensional vector. The second is LBP histogram features, computed using 8 neighboring points at radius 1 with the uniform method. Rather than computing a single histogram over the entire image, the face was divided into a grid of equal-sized cells and a separate LBP histogram was computed for each cell. These histograms were concatenated to

form the final feature vector. This grid-based approach preserves spatial information about where texture patterns occur on the face, which a single global histogram would lose.

An experiment was conducted to determine the optimal grid size by testing 4x4, 6x6, 8x8, and 10x10 grids and measuring SVM accuracy for each.

The optimal grid size was found to be 8x8, achieving 91.0% accuracy. Smaller grids such as 4x4 captured less spatial detail, while larger grids such as 10x10 showed slight accuracy reduction, likely due to cells becoming too small to contain meaningful texture patterns.



Grid=4x4: 87.5%
Grid=6x6: 90.5%
Grid=8x8: 91.0%
Grid=10x10: 90.5%

2.3 Classification

Four model configurations were trained and tested: SVM on raw pixels, SVM on LBP features, MLP on raw pixels, and MLP on LBP features. For SVM, several combinations of C and gamma

were tested to find the best hyperparameters. For MLP, multiple hidden layer architectures and learning rates were evaluated.

C	gamma	Acc Raw	Acc LBP
0.1	0.001	0.0%	0.0%
1.0	0.001	0.0%	80.0%
5.0	0.001	92.0%	91.0%
5.0	0.01	95.5%	90.0%
5.0	0.1	42.0%	88.5%
10.0	0.001	94.5%	90.5%

The SVM was evaluated across six combinations of C and gamma. For raw pixel features, the best configuration was C=5.0 and gamma=0.01, achieving 95.5% accuracy. Low values of C such as 0.1 and 1.0 performed poorly on raw pixels, failing to find a useful decision boundary in the high-dimensional 2500-feature space. Very high gamma values such as 0.1 caused overfitting, dropping accuracy to 42.0%. For LBP features, performance was more stable across C values, with C=5.0 and gamma=0.001 achieving 91.0%. The consistency of LBP results across different C values suggests that the grid-based LBP features produce a more separable feature space that is less sensitive to the regularization setting.

Architecture	LR	Acc Raw	Acc LBP
(64,)	0.001	93.0%	90.0%
(64,)	0.01	54.0%	87.5%
(128,)	0.001	94.5%	92.0%
(128,)	0.01	94.5%	91.0%
(128, 64)	0.001	91.5%	86.0%
(128, 64)	0.01	83.5%	79.0%
(128, 64, 128)	0.001	90.0%	73.5%
(128, 64, 128)	0.01	63.0%	69.5%
(256, 128, 64)	0.001	89.0%	77.5%
(256, 128, 64)	0.01	84.0%	70.0%

The MLP was evaluated across five architectures and two learning rates. The best configuration was a single hidden layer of 128 neurons with learning rate 0.001, achieving 94.5% accuracy on raw pixels and 92.0% on LBP features. Simpler architectures such as (64,) performed comparably on raw pixels but slightly worse on LBP. Deeper networks such as (128, 64, 128) and (256, 128, 64) showed reduced performance, suggesting that added depth causes overfitting given the small dataset size of 200 training samples. A lower learning rate of 0.001 consistently outperformed 0.01, indicating that smaller update steps lead to better convergence on this problem.

3. Results and Analysis

3.1 Accuracy

SVM (raw): 96.0%

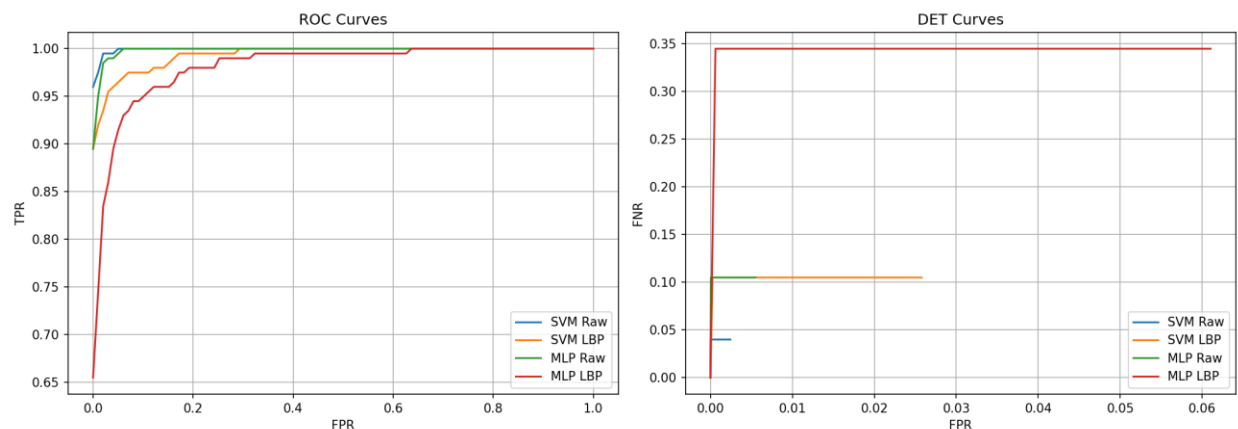
SVM (LBP): 91.0%

MLP (raw): 90.0%

MLP (LBP): 73.5%

The SVM with raw pixel features achieved the highest accuracy at 96.0%, followed by SVM with LBP features at 91.0% and MLP with raw pixels at 90.0%. Contrary to the typical expectation, LBP features did not improve performance over raw pixels for either classifier on this dataset. SVM accuracy dropped by 5.0 percentage points when switching from raw pixels to LBP, and MLP accuracy dropped by 16.5 percentage points. This outcome is likely explained by the small and controlled nature of the AT&T dataset, where images are already well-aligned and captured under consistent conditions. In such cases, raw pixel intensities carry sufficient discriminative information and the texture abstraction provided by LBP removes some of that detail rather than adding robustness.

3.2 ROC and DET Curves



The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR). A curve closer to the top-left corner indicates better performance. The DET curve plots the False Negative Rate (FNR) against the FPR, where curves closer to the bottom-left represent a better tradeoff between false acceptance and false rejection.

The SVM with raw pixel features produced the best ROC and DET curves, consistent with its highest accuracy of 96.0%. On the ROC curve, it sits closest to the top-left corner, indicating the highest true positive rate at any given false positive rate. The MLP with LBP features showed the weakest curves overall, consistent with its lowest accuracy of 73.5%. Notably, LBP features did not improve the curves for either classifier compared to raw pixels. The SVM curves show a clear separation between raw and LBP inputs, while the MLP curves show a similar pattern. This reinforces the conclusion that for this dataset, raw pixel intensities provide a more discriminative input than LBP histogram features.

4. Conclusion

This project compared four face recognition configurations on the AT&T database using SVM and MLP classifiers with raw pixel and LBP histogram features. The results show that SVM outperforms MLP on this dataset, and that raw pixel features outperformed LBP features for both classifiers.

The best overall result was achieved by SVM with raw pixel features at 96.0% accuracy. The limited training set size of 200 samples across 40 classes likely constrains MLP performance, as neural networks generally require more data to generalize effectively. SVM is better suited to small, high-dimensional classification problems, which explains its stronger performance here.

The underperformance of LBP features on this dataset suggests that the texture abstraction it provides is more beneficial in unconstrained settings with variable lighting and pose, rather than controlled datasets like AT&T where raw pixel alignment already carries strong discriminative information.

LLBP reduces the input from 2500 dimensions to 64, making it computationally efficient, though in this case it did not improve discriminability over raw pixels due to the controlled nature of the dataset.

References

[1] Lecture notes and Labs